



MODULE 4

DATA COLLECTION INTERPRETATION AND REPORTING

DATA COLLECTION

- ▶ The process of collecting data from the samples.
- ▶ It begins after defining the research problem & research design.

TYPES OF DATA

Primary data : The data collected fresh and for the first time. The data would be original.

Secondary data: The data that has already been collected by someone and which have already been passed through the statistical process.



PRIMARY VS SECONDARY DATA



PRIMARY DATA	SECONDARY DATA
Real time data	Past Data
Specific to the researchers needs.	Not specific to the researchers needs.
Expensive	Low cost or Free
High level of control on data quality.	Lacks control over data quality.
More time consuming	Less time consuming
More capable to solve specific problem.	Less capable to solve a specific problem.
Owner of data, data can be hidden from competitors.	Competitors also have access to data.

Primary Data Collection Methods



The diagram features a central blue circle with a white border containing the text 'Primary Data Collection Methods'. Four lines radiate from the right side of this circle, each ending in a small colored circle (yellow, orange, pink, and teal). These lines connect to four horizontal chevron-shaped boxes pointing to the right. Each box contains an icon in a diamond shape and a text label. The boxes are colored green, orange, purple, and teal respectively. The icons represent an interview, questionnaires/surveys, observation, and schedules.



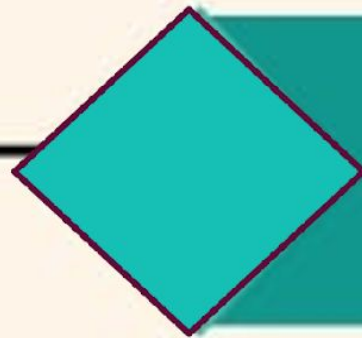
Interview



Questionnaires
and Surveys



Observation



Schedules

OBSERVATION METHOD

- ▶ The information is collected by investigator's own direct observation.
- ▶ The Respondent is not involved in the process.
- ▶ Observation becomes a scientific tool, when it serves a formulated research purpose.

Things to keep in mind while planning an observation.

1. Need to be planned systematically.
2. What is to be observed?
3. How the observations should be recorded?
4. How the accuracy of observation can be ensured?



STRUCTURED VS UNSTRUCTURED OBSERVATION

STRUCTURED OBSERVATION

- ▶ Also called Systematic Observation.
- ▶ The researcher specifies in detail what is to be observed and how the measurements are to be recorded.
- ▶ It defines all the units of observation in advance.
E.g. An auditor performing inventory analysis in a store.

UNSTRUCTURED OBSERVATION

- ▶ The units of observation are not predefined.
- ▶ It monitors all the aspects of the phenomenon that seems relevant to the problem in hand.
E.g. Observing children playing with new toys.

OBSERVATION METHOD PROS AND CONS

MERITS	DEMERITS
Subjective bias is eliminated if done accurately.	It is an expensive method.
Not complicated by the past behavior or future intentions.	The information provided by this method is very limited
It is independent of respondent's willingness to respond.	Unforeseen factors may interfere with the observational task.
Relatively less demanding of active cooperation from respondents.	People are rarely accessible to direct observation

BASED ON OBSERVER PARTICIPATION

- ▶ **PARTICIPANT OBSERVATION** : If the observer observes by making himself, a member of the group he is observing.
- ▶ **NON-PARTICIPANT OBSERVATION** : The observer observes without any attempt on his part to experience through participation what others feel.
- ▶ **DISGUISED OBSERVATION**: The observer is observing in such a manner that his presence may be unknown to the people he is observing.



MERITS & DEMERITS OF PARTICIPANT OBSERVATION

MERITS	DEMERITS
• Researcher is enabled to record the natural behavior of the group directly.	• Observers' presence may create artificial situation.
• Researcher can even verify the truth of statements made by informants.	• Time consuming and challenging to collect data while participating.
• Gets more of direct and information. • Gather more information easily.	• May lose the objectivity to the extent he participates emotionally.

BASED ON OBSERVATION SETTING

CONTROLLED OBSERVATION: Observation takes place according to definite pre-arranged plans.

- It takes place in a laboratory or under controlled conditions.
- We use mechanical instruments as aids to accuracy and standardization.
- These observations supply formalized data.

NON-CONTROLLED OBSERVATION: The observation takes place in the natural setting.

- It gives a spontaneous picture of life and persons.
- It tends to supply naturalness and completeness of behavior.
- Allows sufficient time for observing.

INTERVIEW METHOD FOR DATA COLLECTION



PERSONAL INTERVIEW



TELEPHONIC INTERVIEW

INTERVIEW METHOD - PERSONAL

PERSONAL INTERVIEW : It is a face-to-face meeting or two-way communication between the interviewer and the interviewee.

BASIC PRINCIPLES OF INTERVIEWING

1. Interviewers should be carefully selected and trained.
2. Interviewer need to be honest, sincere, impartial.
3. Interviewer should have good practical experience.
4. Occasional field checks.
5. Appropriate actions, if the respondents refuse to cooperate.
6. Should create friendly atmosphere of trust and confidence.
7. The questions must be asked properly and intelligently.
8. The responses must be recorded accurately and completely.
9. The interviewer must answer legitimate questions by the respondent.
10. The interviewer must clear the doubts by the respondent.
11. The interviewer should not show surprise or disapproval of the responses.
12. Should discouraging irrelevant conversation.



Direct Personal Investigation

VS

Indirect Oral Investigation

DIRECT PERSONAL INVESTIGATION

- ▶ The interviewer collects the information personally from the sources concerned by meeting in person.
- ▶ It involves spot enquiry.
- ▶ Provides the investigator an opportunity to explain the study and objectives.
- ▶ The investigator can also ask the respondent to expand the answers essential for the investigation.
- ▶ The respondents might also provide supplementary information to the interviewer.

INDIRECT ORAL INVESTIGATION

- ▶ The interviewer must approach a third person who has knowledge about the problem under investigation.
- ▶ It is used when the area of investigation is large.
- ▶ Used when respondents feel reluctant in parting information known to them.
- ▶ The investigator prepares a list of questions related to the investigation, and then asks from different persons and records their answers.
- ▶ The person from whom these questions are asked is known as a **witness**.

PERSONAL INTERVIEW TYPES

STRUCTURED INTERVIEW

- ▶ Uses a set of predetermined questions & highly standardized techniques of recording.
- ▶ Here the interviewer follows a rigid procedure of, asking questions.
- ▶ Interviewer cannot skip or alter the sequence of questions.

UNSTRUCTURED INTERVIEWS

- ▶ They follow a more flexible approach with no pre-determined questions and standardized recording techniques.
- ▶ The interviewer has the freedom to ask, supplementary questions or omit certain questions if required.
- ▶ He may even change the sequence of questions.
- ▶ Its needs deep knowledge and greater skill on the part of the interviewer.

PERSONAL INTERVIEW TYPES

FOCUSSED INTERVIEW

- It focuses on the given experience of the respondent and its effects.
- Interviewer has the freedom to decide the sequence of questions.
- He also has the freedom to explore reasons and motives behind the responses.
- The interview aims to confine the respondent to a discussion that is of concern.

CLINICAL INTERVIEW

- It is concerned with broad underlying feelings wrt to individual's life experience.
- The method of obtaining information is left to the interviewer's discretion.

NON-DIRECTIVE INTERVIEW

The interviewer's role is to encourage the respondent to talk on the given topic.

There involves minimum questioning.

PERSONAL INTERVIEW

MERITS	DEMERITS
• Can obtain more information in greater depth.	• It is a very expensive method, specially when the sample is very large.
• Offers greater flexibility, the questions can be restructured as required.	• Possibility of the bias by the interviewer as well as the respondent.
• The interviewer can collect supplementary information about the respondent's personal characteristics.	• Relatively more-time-consuming, specially when the sample is large.
• Interviewer can control which persons will answer the questions.	• Certain types of respondents (high profile), may not be easily approachable.

TELEPHONIC INTERVIEW

- ▶ This method of collecting information involves contacting respondents over telephone.
- ▶ It is not a very widely used method but plays important part in industrial surveys.



TELEPHONIC INTERVIEW

MERITS	DEMERITS
• It is more flexible, faster, cheaper in comparison to other method.	• Little time is given to respondents for answers.
• Recall is easy, callbacks are simple and economical.	• Interview period is not likely to exceed five minutes in most cases.
• There is a higher rate of response, and respondents are easy to access.	• Surveys are restricted to respondents who have telephone facilities.
• Interviewer can explain requirements more easily, wider distribution of sample is possible.	• Possibility of the bias of the interviewer is relatively more.
• No field staff is required to recorded the responses.	• Questions must be short and to the point.

QUESTIONNAIRE METHOD OF DATA COLLECTION



DATA COLLECTION - QUESTIONNAIRES

- ▶ Used for big enquiries.
- ▶ A questionnaire is sent to the respondents with a request to answer and return.
- ▶ A questionnaire consists of several questions printed or typed.
- ▶ The respondents must answer the questions on their own.
- ▶ Quite often questionnaire is considered as the heart of a survey operation.
- ▶ It is used by private individuals, research workers, private and public organizations and governments.



QUESTIONNAIR METHOD

MERITS	DEMERITS
• Low cost even when the universe is large and is widely spread geographically.	• Low rate of return of the duly filled in questionnaires. It can be used only when respondents are cooperating.
• It is free from the bias of the interviewer, and answers are in respondents own words.	• The control over questionnaire may be lost once it is sent.
• Respondents have adequate time to give out the answers.	• Possibility of ambiguous replies, Omission of replies altogether, leads to bias.
• Respondents, who are not easily approachable, can be reached conveniently.	• This method of data collection is the slowest of all.
• Large samples can be made use of and thus the results can be made more dependable and reliable	• It is difficult to know whether respondents are truly representative.

STRUCTURED QUESTIONNAIRE

1. It has concrete and pre-determined questions.
2. The wording and the order of questions is same for all respondents.
3. The responses for question will be closed type.
4. Questions can have alternative replies.
5. They are inexpensive to analyze.

Eg: Do you like the college library services?

☐ YES ☐ NO

UNSTRUCTURED QUESTIONNAIRE

1. Question formulation is largely the responsibility of the researcher.
2. Uses open ended Question with no choices.
3. These questions Invite free responses.
4. Here the replies are to be taken down in the respondent's own words .

Eg: What are the services and facilities you expect from college library?

Response: _____



QUESTION SEQUENCE

- ▶ Questions that are easiest to answer must be put in the beginning.
- ▶ The opening questions should arouse human interest.
- ▶ Questions concerning the research should follow the opening sequence.
- ▶ Question-sequence goes from the general to the more specific.

Avoided questions like:

1. Questions that put great strain on the respondent.
2. Questions of a personal character.
3. Questions related to personal wealth.



Start with simple and non-threatening questions



Group related questions together



Use logical progression
(General to Specific)



Avoid bias and leading questions

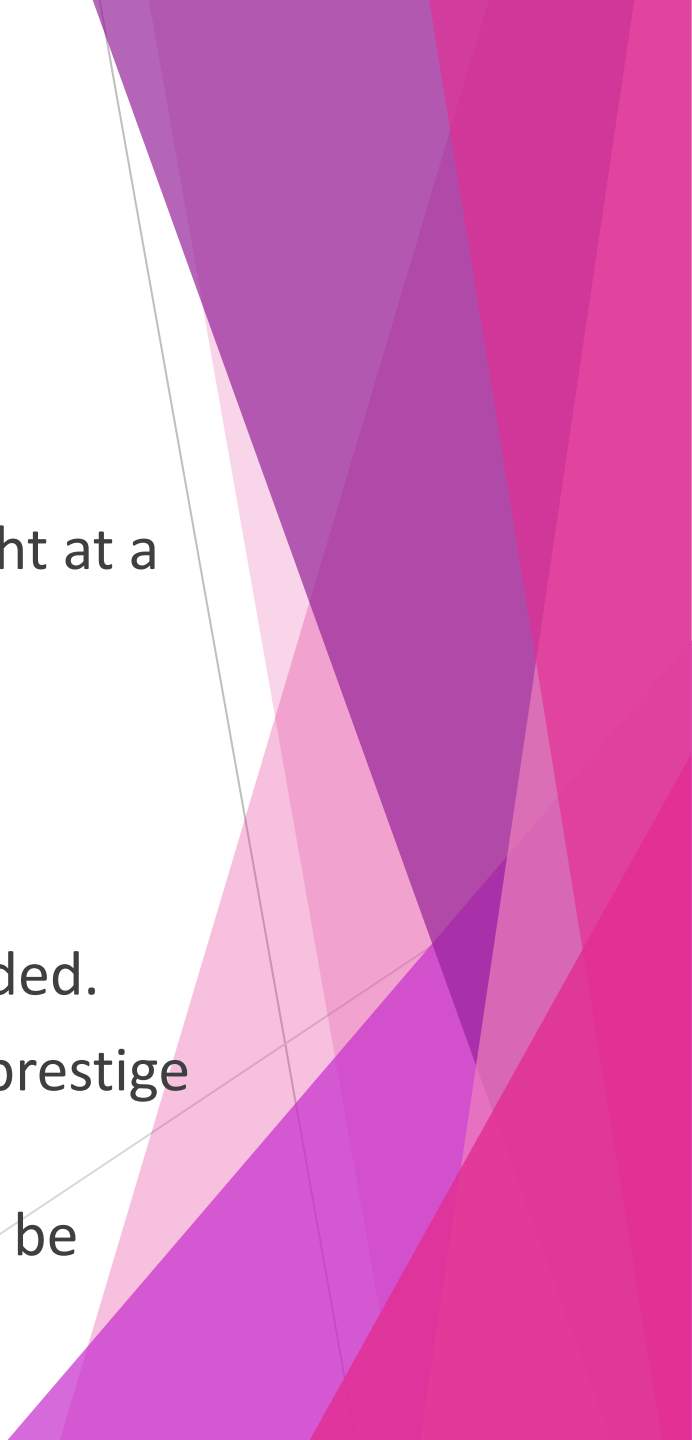


Consider the flow of the overall survey



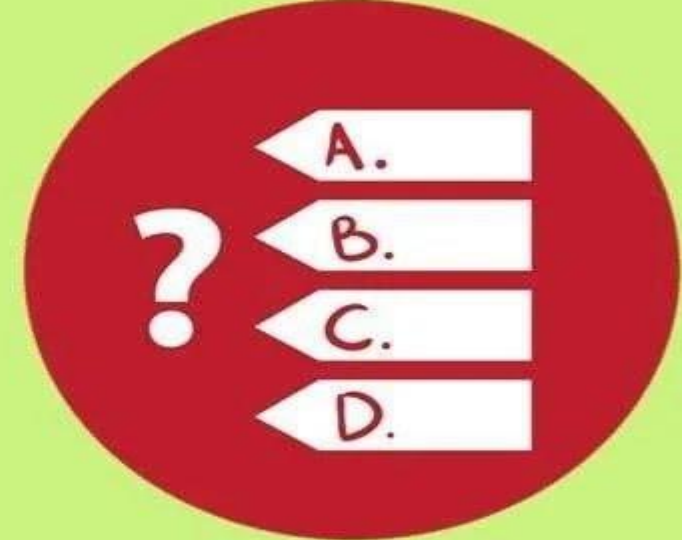


QUESTION FORMULATION AND WORDING

1. Questions must be very clear.
 2. Questions should also be impartial.
 3. Question should be easy to understand.
 4. Questions must be simple & convey only one thought at a time.
 5. Simple and familiar words must be chosen to frame questions.
 6. Words with ambiguous meanings must be avoided.
 7. Words with emotional connotations should be avoided.
 8. Be Cautious in using phrases that reflect upon the prestige of the respondent.
 9. Question wording and formulation is an art and can be learnt only by practice.
- 



VS



OPEN-END QUESTIONS

1. Respondent must answer in his own words.
2. Takes time to answer.
3. Open-ended questions are more difficult to analyze.
4. Appropriate for complex issues.
5. Respondents have an opportunity to express their opinions.

MULTIPLE CHOICE QUESTIONS

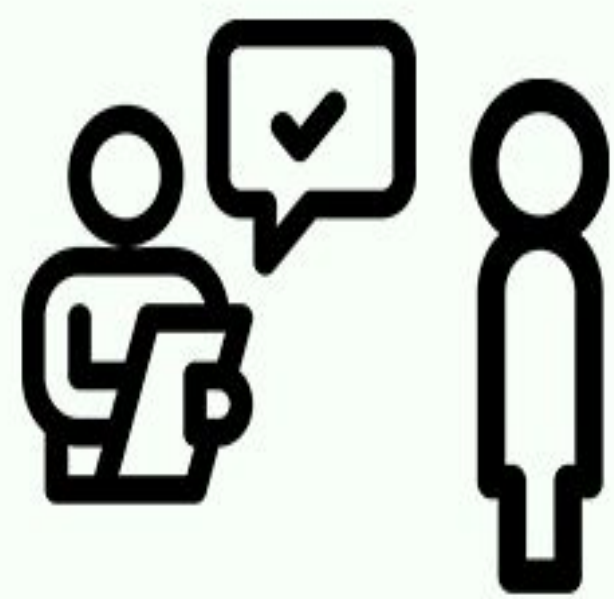
1. Respondent selects one of the alternative possible answers .
2. Closed Questions :Question with only two possible answers ('Yes' or 'No')
3. They can be easily handled, Simple and quick to answer
4. Forces an opinion on an issue about which the respondent has no opinion.

ESSENTIALS OF A GOOD QUESTIONNAIRE

1. The size of the questionnaire should be kept minimum.
2. Questions should proceed in logical sequence, (easy to more difficult ones).
3. Personal and intimate questions should be left to the end.
4. Technical terms and vague expressions should be avoided in a questionnaire.
5. Questions may be dichotomous (yes or no answers), multiple choice (alternative answers listed) or open-ended.

ESSENTIALS OF A GOOD QUESTIONNAIRE

6. There should be some control questions that ensure the reliability of the respondent.
7. Questions affecting the sentiments of respondents should be avoided.
8. Adequate space for answers should be provided in the questionnaire.
9. There should always be provision for indicating uncertainty. (“do not know,” “no preference”).
10. Brief directions about filling up the questionnaire should be given in the questionnaire.



COLLECTION OF DATA THROUGH SCHEDULES

- ▶ **Schedules:** They are proforma containing a set of questions.
- ▶ Schedules are filled in by the enumerators who are specially appointed for the purpose.
- ▶ The enumerators go to respondents and record the replies.
- ▶ Enumerators explain the aims and objects of the investigation.
- ▶ The enumerators should be trained
- ▶ Enumerators be capable of cross examining the truth.
- ▶ They should be honest, sincere, hardworking and should have patience.
- ▶ Schedules are very useful in extensive enquiries.
- ▶ Population census all over the world is conducted through this method.

COLLECTION OF SECONDARY DATA

SECONDARY DATA: The data that is already available. The data which has already been collected and analyzed by someone.

SOURCES OF PUBLISHED DATA

1. Various publications of the central, state and local governments.
2. Publications of foreign governments or of international bodies.
3. Technical and trade journals, books, magazines and newspapers.
4. Reports and publications of various associations like industry, banks, stock exchanges.
5. Reports by research scholars, Public records, historical documents.

SOURCES OF UNPUBLISHED DATA

1. Diary write ups, Letters
2. Unpublished biographies and autobiographies
3. Scholars and research workers, trade associations, labor bureaus.
4. Public/ private individuals and organizations.



CHARACTERISTICS OF SECONDARY DATA

SECONDARY DATA: The data that is already available. The data which has already been collected and analyzed by someone.

RELIABILITY OF DATA

1. Who collected the data?
2. What were the sources of data?
3. Were they collected using proper methods, was there any bias ?
4. At what time were they collected?
5. What level of accuracy was desired? Was it achieved ?

CHARACTERISTICS OF SECONDARY DATA

SUITABILITY OF DATA

- ▶ The data that are suitable for one enquiry may not be suitable for another enquiry, nature of original enquiry might be different.

ADEQUACY OF DATA

- ▶ Data is inadequate if it does not meet the desired level of accuracy.
- ▶ Data is inadequate if it is related to an area which may be either narrow or wider than the area of the present enquiry.
- ▶ Researcher should not use secondary data without proper validation.

DATA INTERPRETATION AND REPORT WRITING



INTERPRETATION OF DATA

- ▶ Interpretation refers to the task of drawing inferences from the collected facts after an analytical or experimental study.
- ▶ Interpretation is concerned with relationships within the collected data.
- ▶ Need to be done carefully, to avoid misleading conclusions.

The task of interpretation has two major aspects

1. Establish continuity in research by linking the results to other studies.
2. Establishment of some explanatory concepts.

WHY INTERPRETATION?

- ▶ The **usefulness and utility** of research findings lie in proper interpretation.
- ▶ Interpretation gives the researcher a good **understanding of the abstract principle** that works beneath the findings.
- ▶ If hypotheses are tested and upheld several times, the researcher may **arrive at generalizations that leads to establishment of explanatory concepts** .
- ▶ If the researcher has no hypothesis to start with, he will try to **explain the findings based on some theory**. (as in Exploratory study)

TECHNIQUE OF INTERPRETATION

Interpretation is an art that one learns through practice and experience.

The steps in interpretation:

1. Researcher must give reasonable explanations for the relations derived.
2. Additional information, collected during the study, must be considered.
3. Before the final interpretation it is advisable to consult someone having insight into the study.
4. The person need to be frank and honest enough to point out omissions and errors.
5. Must consider all relevant factors affecting the problem to avoid false generalization. (Never hurry)



Report Writing



STEPS IN WRITING REPORT

The usual steps involved in writing report are:

1. Logical analysis of the subject-matter
2. Preparation of the final outline
3. Preparation of the rough draft
4. Rewriting and polishing
5. Preparation of the final bibliography
6. Writing the final draft



1. LOGICAL ANALYSIS OF THE SUBJECT-MATTER

Logical analysis is concerned with the development of a subject logically or chronologically.

- ▶ **Logical Development:** The logical development is made based on mental connections and associations between the one thing and another by means of analysis.
- ▶ **Chronological Development:** It is based on a connection or sequence in time or occurrence.

2. PREPARATION OF THE FINAL OUTLINE

- ▶ Outlines are the framework upon which long written works are constructed.
- ▶ They are an aid to the logical organization of the material.
- ▶ They act as a reminder of the points to be stressed in the report.

3. PREPARATION OF THE ROUGH DRAFT

- ▶ The researcher roughly writes down what he has done in the context of his research study.
- ▶ He will write down the procedure adopted by him for
 - ❖ Sampling
 - ❖ Data collection.
 - ❖ Techniques for Data Analysis.
- ▶ Limitations faced by the researcher.
- ▶ Broad findings and generalizations with respect to the study.
- ▶ Various suggestions he wants to offer regarding the problem concerned.

4. REWRITING AND POLISHING

- ▶ This is the most difficult part of all formal writing and requires more time.
- ▶ Check the report for weaknesses in the presentation.
- ▶ He should check the mechanics of writing—grammar, spelling and usage.

5. PREPARATION OF THE FINAL BIBLIOGRAPHY

- ▶ **Bibliography** is a list of books or works referenced by the researcher.
- ▶ The bibliography should be arranged alphabetically.
 1. **Part1** : Names of books and pamphlets.
 2. **Part 2** : Names of magazine and newspaper articles.

BIBLIOGRAPHY ENTRY FORMAT

The bibliography entries can have more than one acceptable form, but importantly it must remain consistent.

FOR BOOKS AND PAMPHLETS

1. Name of author, **last name first**.
2. Title of book, indicate italics.
3. Place, publisher
4. Date of publication.
5. Number of volumes.

Example

Kothari, C.R., *Quantitative Techniques*, New Delhi, Vikas Publishing House Pvt. Ltd., 1978.

FOR MAGAZINES AND NEWSPAPERS

1. Name of the author, **last name first**.
2. Title of article, in quotation marks.
3. Name of periodical.
4. The volume number.
5. The date of the issue.
6. The pagination.

Example

Robert V. Roosa, "Coping with Short-term International Money Flows", *The Banker*, London, September 1971, p. 995

6. WRITING THE FINAL DRAFT

- ▶ The final draft is the most effective in communicating the research findings to others.
- ▶ It should be written in concise and objective style and in simple language.
- ▶ The report should avoid vague expressions such as “it seems”, “there may be” etc.
- ▶ The researcher must avoid abstract terminology (hard to understand).
- ▶ A research report should not be dull, It needs to maintain interest for the reader.
- ▶ The report must have originality.

LAYOUT OF THE RESEARCH REPORT

1. PRELIMINARY PAGES

- ▶ Title and date
- ▶ Acknowledgements.
- ▶ Table of contents (Index)
- ▶ List of tables and illustrations.

2. MAIN TEXT

It provides the complete outline of the research report. Each main section of the report should begin on a new page.

- ▶ Introduction

SECTIONS OF MAIN TEXT

1. INTRODUCTION

- ▶ It Introduces the research project to the readers.
- ▶ It should contain a clear statement of the objectives of research.
- ▶ The hypotheses of study, if any.
- ▶ The definitions of the major concepts employed in the study.
- ▶ The methodology adopted in conducting the study.
- ▶ The scope of the study.
- ▶ Should narrate the various limitations, under which the research project was completed.

SECTIONS OF MAIN TEXT

2. STATEMENT OF FINDINGS AND RECOMMENDATIONS

- ▶ The statement of findings and recommendations must be stated in a non-technical and summarized manner so that it can be easily understood by all.

3. RESULTS

- ▶ Presentation of the findings of the study, (supporting data as tables and charts.
- ▶ Should contain statistical summaries.
- ▶ Results should be presented in a logical sequence.
- ▶ Should include all relevant results.
- ▶ Clearly state the problem concerned and how he worked on it.
- ▶ The conclusions from the study, and the reasoning behind the conclusions.

SECTIONS OF MAIN TEXT

4. IMPLICATIONS OF THE RESULTS

The conclusion drawn should be related to the hypotheses stated.

- ▶ Inferences drawn from the present study may apply in similar circumstances.
- ▶ The conditions of the present study may limit the extent of legitimate generalizations of the inferences.
- ▶ The relevant questions that remain unanswered or new questions raised by the study with suggestions on the kind of research that may apply.

5. SUMMARY

- ▶ The researcher should conclude the research report with a summary. The summary extracts and highlights the most important points from a larger body of work.

TYPES OF REPORTS

Research reports vary greatly in length and type.

- ▶ **Business firms:** Reports in the letter form, just one or two pages in length.
- ▶ **Financial institutions:** Short balance-sheet type of tabulation for their annual reports
- ▶ **Mathematicians:** Present the results in algebraic notations.
- ▶ **Chemists:** Present the results in symbols and formulae.
- ▶ **Students of literature:** long reports presenting the critical analysis of some writer.
- ▶ **Clinical psychologists:** case-history form.
- ▶ **News items** in the daily papers are also forms of report writing.
- ▶ **Book-reviews:** Analyze the content of the book and report on the authors' intentions, his language, his style etc.
- ▶ **Ph.D. theses** report-writing, usually done by students in academic institutions.

TECHNICAL VS POPULAR REPORT

TECHNICAL REPORT

- ▶ It describes the process, progress, or results of scientific research.
- ▶ It emphasizes on the assumptions made during study.
- ▶ It also include recommendations and conclusions of the research.
- ▶ It is very lengthy with multiple identifiable sections.

POPULAR REPORT BUSINESS REPORT

- ▶ An approach of presenting financial and non-financial information to non-professional users (e.g., citizens) in a concise, and easy-to-read format.
- ▶ Aims at simplicity and attractiveness and policy implications.
- ▶ Involves minimal of technical details, and liberal use of charts and diagrams.
- ▶ Attractive layout along with large prints, many subheadings, even an occasional cartoon.

TECHNICAL REPORT

OUTLINE OF A TECHNICAL REPORT

1. **Summary of results:** Review of the main findings.
2. **Nature of the study:** General objectives of study, formulation of the problem in operational terms, the working hypothesis, the type of analysis and data required, etc.
3. **Methods employed:** Specific methods used in the study and their limitations.
4. **Data:** Data collected, their sources, methods employed, characteristics and limitations.
5. **Analysis of data and presentation of findings:** Presentation of the findings in the form of tables and charts. (Main Body)
6. **Conclusions:** A detailed summary of the findings and the implications drawn from the results.
7. **Bibliography:** Bibliography of various sources consulted.
8. **Technical appendices:** Additional information on all the technical matters.
9. **Index:** Index must be prepared and be given invariably in the report at the end.

POPULAR REPORT

OUTLINE OF A POPULAR REPORT

1. **The findings and their implications:** Emphasis in the report is given on the findings of most practical interest.
2. **Recommendations for action:** Recommendations for action based on the findings of the study.
3. **Objectives of the study:** A general review of the objectives of the project under study.
4. **Methods employed:** Non-technical description of the methods used.
5. **Results:** The results are presented in clear and non-technical terms with liberal use of all sorts of illustrations such as charts, diagrams etc.
6. **Technical appendices:** More detailed information on methods used, forms.

MECHANICS OF WRITING A RESEARCH REPORT

1. SIZE AND PHYSICAL DESIGN

- ▶ The manuscript should be written on unruled paper 8 1/2" × 11" in size.

If written by hand

- ▶ Use black or blue-black ink should be used.
- ▶ A margin of at least 1.5 inch on left, and 0.5 inch on right hand side of paper.
- ▶ There should also be one-inch margins, top and bottom.
- ▶ The paper should be neat and legible.
- ▶ **If typed**, all typing should be double-spaced.

MECHANICS OF WRITING A RESEARCH REPORT

2. PROCEDURE

- Various steps in writing the report should be strictly adhered.

3. LAYOUT

- Layout of the report should be thought of and decided and accordingly adopted based on the objective.

4. TREATMENT OF QUOTATIONS

- Quotations should be placed in quotation marks and double spaced.
- If of considerable length, then it should be single-spaced and indented at least 0.5" to the right of the normal text margin.

MECHANICS OF WRITING A RESEARCH REPORT

5. THE FOOTNOTES

- ▶ Footnotes are notes placed at the bottom of the page in a piece of academic writing and indicated in the text with superscript numbers.
- ▶ These notes serve as references to cited works and offer supplementary information to aid the reader's understanding.
- ▶ Footnotes should be numbered consecutively, usually beginning with 1 in each chapter separately.
- ▶ Footnotes are always typed in single space though they are divided from one another by double space.

MECHANICS OF WRITING RESEARCH REPORT

Reference Format

1. Author's name in normal order
2. Title of work, underlined to indicate italics.
3. Place and date of publication.
4. Number of volume (multivolume reference)
5. Pagination references.

Regarding periodicals reference

Second-hand quotations

1. Original author and title “quoted or cited in,”
2. Second author and work

Case of multiple authorship

1. Name of first author is given and the others are indicated by “et al.” or “and others”.
2. A single page should be referred to as p.
3. More than one page be referred to as pp.
4. If Several pages referred to at a stretch eg: pp. 190ff, (190 and following pgs).
5. If only for page 190 and the following page ‘190f’.
6. Roman numerical is generally used to indicate the number of the volume of a book.

COMMON ABBREVIATIONS IN REPORT-WRITING

1. **anon.**, anonymous
2. **ante.**, before
3. **art.**, article
4. **aug.**, augmented
5. **bk.**, book
6. **bull.**, bulletin
7. **cf.**, compare
8. **ch.**, chapter
9. **col.**, column
10. **diss.**, dissertation
11. **ed.**, editor, edition, edited.
12. **ed. cit.**, edition cited
13. **e.g.** for example
14. **eng.**, enlarged

6. USE OF STATISTICS, CHARTS AND GRAPHS

1. The use of statistics in research reports contributes towards the clarification and simplification of the research results.
2. Statistics are usually presented in the form of tables, charts, bars and line-graphs and pictograms.
3. Statistics , charts and graphs should be self explanatory.
4. Statistical presentations should be neat and attractive.

7. THE FINAL DRAFT

1. Rewriting the rough draft of the report should be done with great care.
2. Are the sentences written in the report clear?
3. Are they grammatically, correct?
4. Do they say what is meant'?
5. Do the various points incorporated in the report fit together logically?
6. Sentences that seem clear to the writer may prove quite confusing to others.

8. BIBLIOGRAPHY

- ▶ Bibliography should be prepared and appended to the research report.

9. PREPARATION OF THE INDEX

- ▶ Index, acts as a good guide, to the reader.
- ▶ Index may be prepared both as subject index and as author index.
- ▶ **SUBJECT INDEX** gives the names of the subject-topics or concepts along with the number of pages on which they have appeared or discussed in the report.
- ▶ **AUTHOR INDEX** gives the similar information regarding the names of authors.
- ▶ The index should always be arranged alphabetically.